

MIRROR PAGE

A comic-strip light-routing puzzle where an unreliable narrator twists the rules. • TGC Game Jam 2026 Proposal

Team / Creator	Sankeerth Nara (Solo)	Email	sankeerthnara@gmail.com
IndieConnect ID	[YOUR_INDIECONNECT_ID]	Public GitHub	github.com/SankeerthNara/TGC-Game-Jam (MIT)
Themes (All 3)	COMIC TWIST LIGHT	Target Playtime	12 – 15 mins first playthrough (20 min hard cap)
Engine & Renderer	Godot 4.7, GDScript only, GL Compatibility Renderer (Web/HTML5 export hosted on itch.io)		

1. PITCH & CORE CONCEPT

Somebody has sabotaged the lights of an old comic studio. You are a small caped comic hero with a torch in a **pitch-dark station** split into **4 levels**, each a freshly generated maze of 8 rooms in a random layout: only the area around you is lit. Explore, find the task consoles, and finish each level's **5 to 8 interactive tasks** (rewire panels, order comic panels, fill speech bubbles, debug code, wire logic gates, align a telescope and more) to fill the **progress bar**. Careful: some tasks trigger a **sabotage countdown**. Fix it in time or lose hearts, and die at zero. Some tasks are light puzzles on comic pages where you drag panels and flip mirrors to steer a beam. Who cut the lights, and why? The narrator has a few twists planned.

2. INTEGRATION OF THE THREE THEMES

Theme	How It Drives Gameplay & Narrative
Comic	The station is a comic-book studio (ink outlines, halftone paper, narrator captions, onomatopoeia) and the light puzzles are comic pages whose panels are movable tiles; gutters act as light refractors.
Light	The whole map is dark by design: only the area around your torch is visible, so light is how you explore. Light is also the puzzle: steering a beam off mirrors onto star targets.
Twist	The light sabotage is the story twist: the saboteur is among the characters and is revealed late. In the puzzle pages the narrator also twists the rules: the hero you light turns out to be the villain, mirrors lie, and gutters bend light 90°. (More twists are planned.)

3. CORE MECHANICS

- Explore the dark:** Smooth 8-direction walking in a seamless tile map, a new random 8-room layout every level. Only the area around the torch is lit. Press M for the station map.
- 26 interactive tasks over 4 levels, one progress bar per level:** comic tasks (order the panels, fill speech bubbles, letter sound effects), college tasks (debug code, logic gates, shelf books), plus wires, fuses, lantern code, torch charge, telescope, ink spills, card swipe, and light-puzzle pages.
- Sabotage and health:** six risky tasks start a timer when completed. A complementary fix console must be completed before it runs out, or the hero loses a heart. Zero hearts: respawn at the checkpoint (the start of the current level). Every level is harder than the last and a timer tracks your time per level and in total.
- Light puzzle pages:** drag panels to swap them (their contents move too), click mirrors to flip them, steer the beam onto every goal. Unlimited undo.
- Beam Tracing:** an instant headless raycast lights targets and flags hazards after every move.
- Frictionless:** [Z] interact/undo, [R] reset, [M] map, Esc leave a task; no softlocks.

4. THE NARRATOR'S RULE TWISTS

Rule	Mechanical Impact	Introduced
Normal	Standard optical reflections: mirrors reflect as drawn; gutters maintain beam angle.	Levels 1, 2, 3
Goal Flip	Narrator caption flips polarity: good stars become toxic; hazards (red X) become the required target.	Levels 3 and 6
Lying Mirrors	Mirrors reflect the exact opposite of their drawn slant (slashes act as backslashes).	Level 4
Gutter Bend	Crossing a panel gutter forces a mandatory 90° clockwise beam deflection on the entry cell.	Levels 5 and 6

5. CONTENT PLAN & GAMEPLAY LOOP (15 MINUTES)

#	Level Name	Core Mechanic / Rule Introduced	Target Time
1	Page One (Library)	Panel swapping across gutters to carry a straight beam to the hero's star.	1.5 min
2	Looking Glass	Combining panel swapping with mirror angle toggles and wall obstacles.	2.5 min
3	The Twist	Goal Flip Twist: Solves normally, narrator flips roles, re-solve page for villain.	2.5 min
4	Liar, Liar	Lying Mirrors: Slashes reflect counter-intuitively; mental inversion puzzle.	2.5 min
5	Gutter Bend	Gutter Bend: Beams turn 90° at boundaries; locked panels limit swap freedom.	3.0 min
6	The Last Page	Grand Finale: Combines lying mirrors, gutter bends, locked tiles, and final flip.	3.0 min

Station loop: read the task list → walk through the dark to the next console → play its mini-game → progress bar fills → narrator reacts. 4 levels x 8 random rooms, 26 tasks (5 are light puzzle pages), up to 4 sabotage events per level, checkpoints between levels, then the ending and a split-time summary.

6. SCOPE BOUNDARIES

In Scope: 4 generated dark levels with torch lighting, 26 mini-game tasks with a progress bar, sabotage timers and hearts, station map, 6 tuned light-puzzle pages, 3 rule mutations, goal flip twist, saboteur story twist, narrator caption UI, menus, end screen, comic sound effects, music, web export.

Out of Scope (Strictly Cut): Level editor, save slots, mobile touch, gamepads, multiplayer, more than 20 rooms.

7. TECHNICAL ARCHITECTURE & WEB FEASIBILITY

- **Engine & Renderer:** Godot 4.7 with GL Compatibility renderer; single-threaded Web export with zero native extensions or GDExtensions.
- **Model-View Decoupling:** pure GDScript puzzle model (PageModel1, BeamSolver) separate from rendering; the station (World) and its levels and tasks are data-driven from a JSON map; each mini-game is its own class. Tested headless.
- **Automated Validation:** Built-in BFS/DFS solver (level_check.gd) verifies every level is 100% solvable within 2–8 optimal moves.
- **Web Export:** game package under 1 MB PCK (build about 38 MB uncompressed, ~8 MB gzip). 2D lights and darkness verified in a browser build.

8. 100-HOUR DEVELOPMENT MILESTONES

Hours	Phase & Deliverables	Current Status
0 – 12	Concept lock, core engine prototype, level format, art research, proposal submission.	COMPLETE
12 – 30	Vertical slice: dark station, tasks and progress bar, comic UI, first art pass, CC0 audio, pages 1–6 playable.	IN PROGRESS
30 – 60	Station polish, more tasks and room props, saboteur twist, sound and music integration, difficulty tuning from playtests.	Planned
60 – 85	Balancing, narrative comedic timing, web playtests, UI juice (panel shake, onomatopoeia).	Planned
85 – 100	Feature freeze at Hour 85. Web export audit, itch.io upload, README and CREDITS audit.	Planned

9. ASSETS & AI DISCLOSURE

- **Asset Rule Compliance:** Zero paid assets. All assets are 100% free, CC0, or open source with full attribution in CREDITS.md:
 - **Fonts:** **Bangers** (SIL OFL 1.1) and **Comic Neue** (SIL OFL 1.1) from Google Fonts.
 - **Audio:** Procedural CC0 synthesized sound effects and jazz loop generated in Python.
 - **Art & Textures:** Vintage newsprint textures, Ben-Day halftone dots, and comic burst badges generated procedurally.
- **AI Collaboration:** All AI usage is documented in docs/AI_USAGE.md across Claude Code (architecture/core logic), Antigravity (UI, art/audio assets, browser QA), and ChatGPT (tests/docs). Core creative direction and design decisions remain human-authored.

10. RISKS & MITIGATIONS

Identified Risk	Proactive Mitigation Strategy
Web Export / Browser Bugs	Tested every milestone via Playwright headless Chromium. Compatibility renderer enforced from Day 1.
Puzzles Unsolvable / Obscure	Automated headless level validator verifies solvability and move counts before authoring.
Scope Overrun	Strict 6-level cap, 15-minute loop target, hard feature freeze at Hour 85.